23 When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A and the voltmeter reads zero.

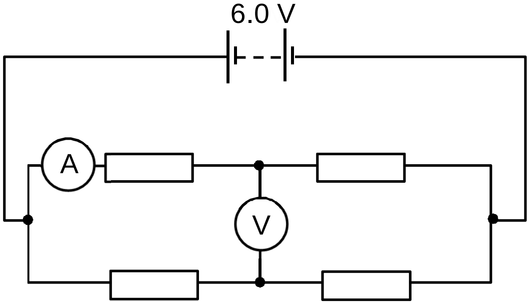

diagram 1

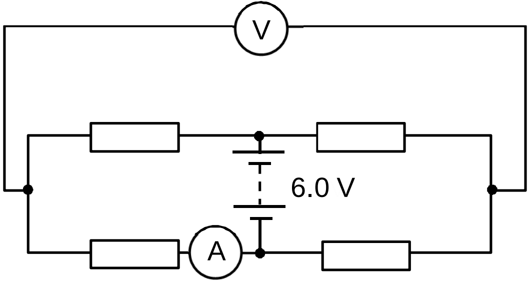

diagram 2

The resistors and meters are reconnected to the supply as shown in diagram 2.

What are the meter readings in diagram 2?

	voltmeter reading / V	ammeter reading / A
A	0	1.0
B	0	0.5
C	3.0	1.0
D	6.0	0.5

Answer: A

Moving the battery source and the meters to the new position does not change the effective layout of the circuit. By circuit analysis the 4 resistors are still connected similar to the first i.e. 2 in series with each other and then these 2 in parallel with the other branch.

Therefore based on the position of the ammeter it is measuring the current in one branch of the circuit hence the current is still 1.0 A and the voltmeter is now measuring across two points in the circuit where the potentials of these two points are the same. So potential difference is 0 V.

OR

Based on diagram 1,

Let the unknown value of the resistors be R.

$$(1.0)(R+R) = 6.0\ \text{V}$$
$$2R = 6.0\ \text{V}$$
$$R = 3.0\ \Omega$$

Based on diagram 2,

$$I(6.0) = 6.0\ \text{V}$$
$$I = 1.0\ \text{A}$$

Since the potential difference is measured after the first resistor in both branches the potentials at these two points are the same. Hence potential difference is 0 V.